National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Linear Circuit Analysis	Course Code:	EE1001
Program:	BS(Electrical Engineering)	Semester:	Spring 2023
Duration:	60 minutes	Total Marks:	30
Paper Date:	28-February-2023	Weightage:	15%
Section:	All	Page(s):	2
Exam Type:	MID – I		

[redacted]

Instruction/Notes:
1. Circuits must be labeled properly, methods/techniques must be indicated and formulas must be written prior to inserting values.
2. Show all steps.
3. Attempt all parts of a question together.

Q.1 [10 marks] [CLO 01]

The current $i(t)$ passing through a 10 mH inductor is given by the waveform shown in Figure 1 below:

$i(t) = \frac{1}{2} L \int_0 v\,dt + i(0)$

$V = 3$

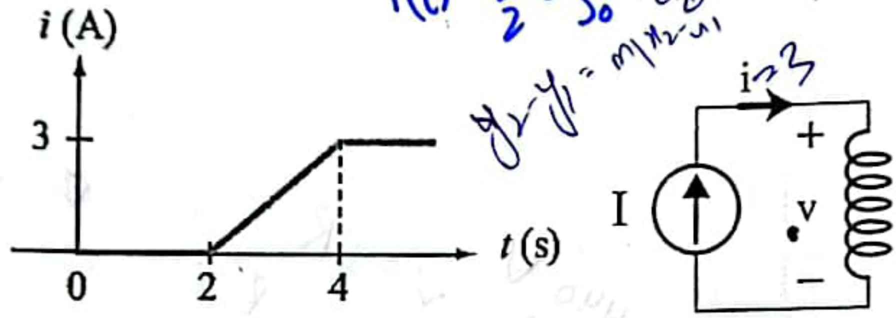

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Figure 1. Waveform and Circuit for the current through the inductor

a) Determine the piece-wise equation of the current from the graph.

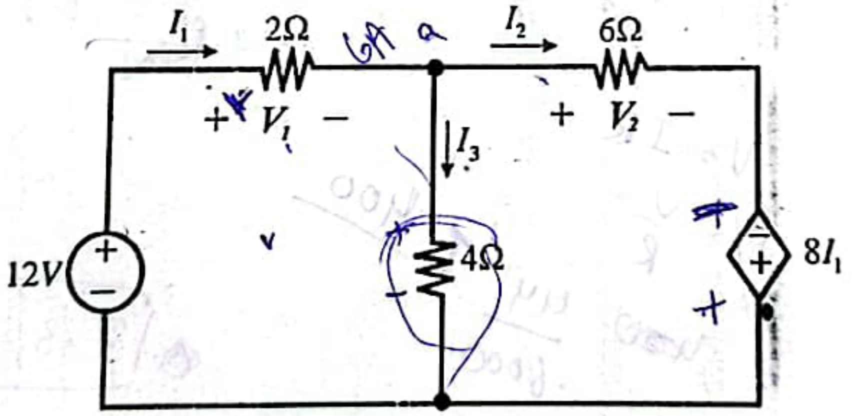
b) Determine corresponding voltage across the inductor and plot it.

$V = Li$

$V = IR$

$I = \frac{V}{R}$

Q.2 For the circuit shown in Figure 2 below: [10 marks] [CLO 02]



$= \frac{12}{2\Omega}$

$\frac{8 \cdot 12}{4} = 6A$

Figure 2. Circuit with Dependent Source

a) Determine the voltages V_1, V_2 and currents I_1, I_2 using Kirchhoff's laws and Ohm's Law.

b) Determine the power associated with the dependent voltage source using passive sign convention.

Q.3 For the circuit shown in Figure 3 below: [10 marks] [CLO 02

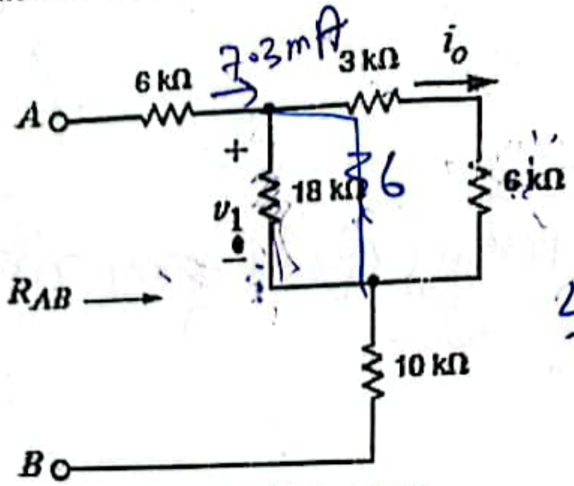


Figure 3. Resistive Circuit for Q3

$V = IR$

$I = \frac{V}{R}$

a) Determine the equivalent resistance between terminals 'A' and 'B'.

b) If a 44V voltage source is connected between terminals 'A' and 'B', use voltage divider rule (VDR) to find the voltage v_1 and current divider rule (CDR) to find the current i_0.

c) Calculate the power absorbed by the 10 kΩ resistor.

$= R_1 + R_2 + R_3$

$= IV_1 + IV_2 + IV_3$

$= I(V_1 + V_2 + V_3$

440 V

$V = IR$

$V = \frac{V}{R}$

7.3×10^{-3} A

$P = VI$

$= I^2R$

$V = \frac{R_j \times i_T}{R_{eq}}$

V_2

$\frac{44}{6000}$

$V = IR$

$I = \frac{V}{R}$

$i_T = 400$

$= \frac{44}{6000}$

$\boxed{7.3 \text{ mV}}$

$V = IR$

7.3×10^{-3} V

$I = \frac{V}{R}$